

KUSH PATEL

Pittsburgh, PA

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🌐 [patelkush1508](https://patelkush1508.github.io)

🔗 [kushp15](https://github.com/kushp15)

Summary

Seeking 2026 AI/ML Engineering internship, combining technical expertise with strong collaboration and problem-solving skills to design scalable AI/ML solutions and drive business impact

Education

Carnegie Mellon University - GPA: 3.6/4.0

May 2027

Masters of Science in Artificial Intelligence Engineering - Information Security (Advanced Study)

Pittsburgh, PA

CourseWork: Adversarial ML, Cloud Security, Advanced Data Networks, Advanced NLP, AI in Information Security

Rutgers University - New Brunswick - GPA: 3.6/4.0

Sept 2022 – Aug 2024

Bachelor of Science in Computer Science, Minor in Business Administration

Piscataway, PA

CourseWork: Artificial Intelligence(A), Deep Learning(A), Computer Security(A), Operating Systems(A), Systems Programming(B+) — **Activities:** RUMAD (**Mentor**), HinduYuva (**Leadership**), Hackathons, Chess Club

Technical Skills

Programming Languages: Python, C, C++, PostgreSQL, React, TypeScript

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Django, Pandas, OpenCV, JAX, Spark, HuggingFace

Tools & Technologies: GitHub, Docker, Kafka, Postman, RESTful APIs, NoSQL

Cloud & Platforms: Google Cloud Platform (GCP), Amazon Web Services (AWS), Linux, Vercel

Experience

Advantest America, Inc

Jun 2024 – Present

Technology Specialist, Developing

South Plainfield, NJ

- Integrated **Model Context Protocol** into the **AI Agent** to generate adaptive, real-time production dashboards, increasing report accuracy and timeliness across global operations **across 5 global plants**
- Designed distributed **ETL pipelines ingesting 150M+ records** via REST APIs into PostgreSQL, ensuring real-time sync and enterprise-grade reliability
- Implemented **CI/CD workflows** with GitHub Actions, **reducing deployment time by 40%** while improving reliability through automated testing
- Built robotic systems using **StandardBots APIs** and TCP/IP protocols to measure PCB thickness with **50–100 mil precision**, improving measurement **accuracy and speed by 90%**.

Advantest America, Inc

Aug 2022 – May 2024

Technology Intern

South Plainfield, NJ

- Managed Azure AD with IAM role assignments, MFA enforcement, and conditional access policies
- Implemented endpoint security using Intune with disk encryption, compliance policies, and application control
- Maintained secure infrastructure across VMware, Windows Server, and Active Directory with patching, DNS security, and GPO hardening
- Led system audits and backup verification, automating asset tracking and compliance reporting

Projects

Computer Vision PCB Inspection Tool @ Advantest | Python, EasyOCR, Linux

2025

- Created a real-time PCB inspection system to automate layer sequence validation using OpenCV and OCR
- Structured logging via **REST APIs to cut inspection latency by 22%**, result real-time data streams and analysis
- Optimized **image processing algorithms** to enhance inspection accuracy and throughput, **saving \$1M annually by reducing manufacturing defects**

Lazy Track | Python, Transformers, Google APIs, Canvas API, SQLite

2025

- Developed an **AI Agent** to monitor emails and extract job-related context using a **Model Context Protocol (MCP)**
- Automated status updates and reminders **Dashboard**, improving job search **efficiency by over 70% with APIs**

Fraud Email Detector Model | Python, PyTorch, GCP, claude

2025

- Built a binary classification model using **TF-IDF and embeddings** to detect fraud in 100K+ real-world email samples
- Achieved 93% accuracy** and reduced false positives, improving inbox safety for end users and reducing phishing risk